Higher National Diploma in Engineering (Civil / Mechanical / Electrical / QS / BSE)

Second Year, First Semester Examination-2017

IT2102-Information Technology II A

Answer only 4 questions **Time: 2 hours**

Q1.

I. Describe the following controls (Marks: 2x5)

a) Text Box
b) Combo Box
c) Check Box
d) Command button
e) Scroll Bar

II. Give two important properties for each controls on the above question and explain their usage. (Marks:2 x 5 Marks)

III. Write syntax of main event procedure for the controls given the above questions. (Marks: 1 x 5 Marks)

Q2.

I. Describe the variable in Visual Basic. (3 Marks)

II. Explain the following scope of the variables:

a. Dim
b. Static
c. Private

(Marks: 2x3 Marks)

III. Give four basic data type of variable in visual basic and describe them. (Marks: 2x4 Marks)

IV. Write suitable variable declaration to store the following information. Also specify whether your variable is a procedure level variable or module level variable.

a. To store a student's marks. Assume mark is a whole number.
b. To store a student's name.
c. To store a student's date of birth.
d. To store the status of registration- Whether the student is registered or not.
e. Suppose a form contains a command button named "cmdGrade". What is the suitable variable declaration so that the variable hold that how many times the button "cmdGrade" is clicked?
f. To store list of 100 whole numbers.
g. To store atmosphere temperature that can be accessed by all procedure in a form module.
h. To store the distance between stars in km in our solar system.

(Marks: 1x8 Marks)

Q3. Conditional Statements and Interface Design

I. Describe the following visual basic statement. Also write the syntax of them:

a. If.....Then.....Else
b. Nested If
c. Select.... Case

(Marks: 3x3 Marks)

II. You are requested to calculate the area of the following shapes

- Circle (area = pi x r x r)
- Rectangle (area = pi x l x w)
- Square (area = pi x l x l)
- Ellipse (area = pi x a x b)

a. Design suitable interface (Form). Your design should allow to select one of the above shapes. (5 Marks)

b. Identify the type of controls you used on your design and explain their usage. (4 Marks)

c. Write necessary event procedures. (7 Marks)

Q4.

I. Compare For....Next Loop and Do While... Loop statements in visual basic.(4 Marks)

II. The following is an event procedure for the command button - *cmdFactorial*. This event procedure is used to calculate the factorial n. Assume the number n is entered in the text box - *txtNumber*.

```
Private sub cmdFactorial
Dim n.1 as integer, fas string, x as integer
n.I=val(txtNumber.Index)
f=1
x=1
Do While x> n.1
f=f*x
x=x+1
End Loop
Print f
End Sub
```

a. Identify the syntax errors on the above event procedure.(6 Marks)
b. After correction of the above event procedure. Can you get the right answer? Explain why.(4 Marks)
c. Rewrite the above program using ***For.. Next*** Loop.(6 Marks)
d. Explain how the following validations can be done on your program.

- The factorial can be done for positive whole number
- The maximum value of ***n*** is 50

(5 Marks)

Q5.

I. What are the advantages of using User Defined Function in programs?

(3 Marks)

II. Write the syntax of User Defined Function.(3 Marks)

III. Consider the following visual basic coding. Where "**txtNumber**" is a text box, and "**Grade**" is a user define function. The "**Grade**" return the grade for the given marks. Write the program for the function - Grade.

visual basic Private sub cmdResult()

Dim N as integer

N=val(txtNumber)

Print Grade(N)

End Sub

(6 Marks)

IV. Compare event procedure and User Define Procedure.(3 Marks)

V. Write user define procedure to swap (exchange) two numbers.(6 Marks)

VI. Create an event procedure to exchange two numbers which are on text boxes using the user define procedure created on the above question.(4 Marks)